

Analog IC Design prelab 02

Common Drain (Source Follower) Amplifier

July 22, 2026

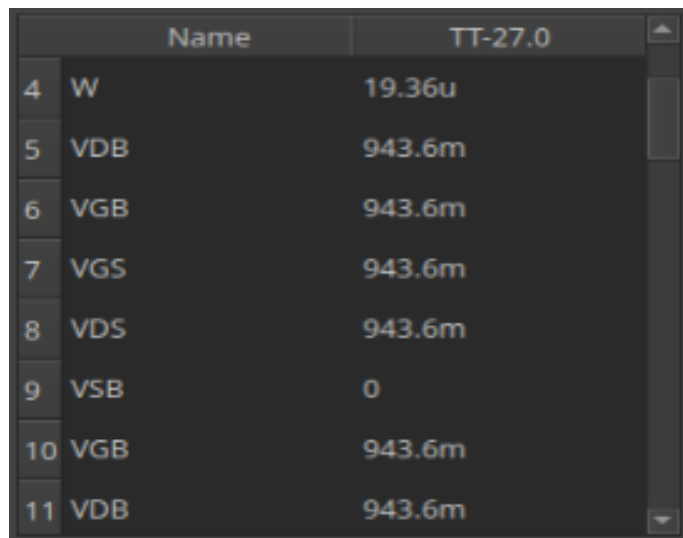
1 Part 1: Device Sizing Using SA

The design spec given was: PMOS input device, $L = 1\ \mu m$, $V^* = 200\ mV$, quiescent input = 0 V, $V_{DD} = 1.8\ V$, $I_D = 10\ \mu A$.

From the square-law relation

$$V^* = \frac{2}{g_m/I_D} \Rightarrow \frac{g_m}{I_D} = \frac{2}{V^*} = \frac{2}{0.2} = 10\ S/A$$

$$g_m = I_D \cdot (g_m/I_D) = 10\ \mu A \times 10 = 100\ \mu A/V$$



	Name	TT-27.0
4	W	19.36u
5	VDB	943.6m
6	VGB	943.6m
7	VGS	943.6m
8	VDS	943.6m
9	VSB	0
10	VGB	943.6m
11	VDB	943.6m

Figure 1: SA output – bias voltages and resulting W .

	Name	TT-27.0
15	gm/gds	423.9
16	VA	42.86
17	ID/W	516.5m
18	gm/W	5.109
19	AREA	19.36p
20	gm	98.91u
21	gmb	46.78u
22	gds	233.3n

Figure 2: SA output – derived small-signal parameters (g_m/g_{ds} , V_A , I_D/W , g_m/W , area, etc).

So SA gives $W = 19.36 \mu m$ (with $L = 1 \mu m$, so $1 \times 19.36 \mu m / 1 \mu m$, matches what's on the schematic).

Quantity	From SA table	Check	Match?
I_D/W	516.5 mA/A ($\mu A/\mu m$)	$10 \mu A / 19.36 \mu m = 516.5 m$	yes
g_m/W	5.109	$98.91 \mu A/V \div 19.36 \mu m = 5.108$	yes
$V_A = I_D/g_{ds}$	42.86 V	$10 \mu A / 233.3 nS = 42.86 V$	yes
Area	19.36 (p)	$W \cdot L = 19.36 \mu m \times 1 \mu m$	yes

Table 1: Consistency check on the SA numbers.

2 Part 2: CD Amplifier

2.1 OP (Operating Point) Analysis

I built the CD amp with the PMOS from Part 1 ($W = 19.36 \mu m$, $L = 1 \mu m$), an ideal $10 \mu A$ current source load connected at the source (with source tied to body), $C_L = 2 pF$, $R_{sig} = 2 M\Omega$, and DC input = 0V.

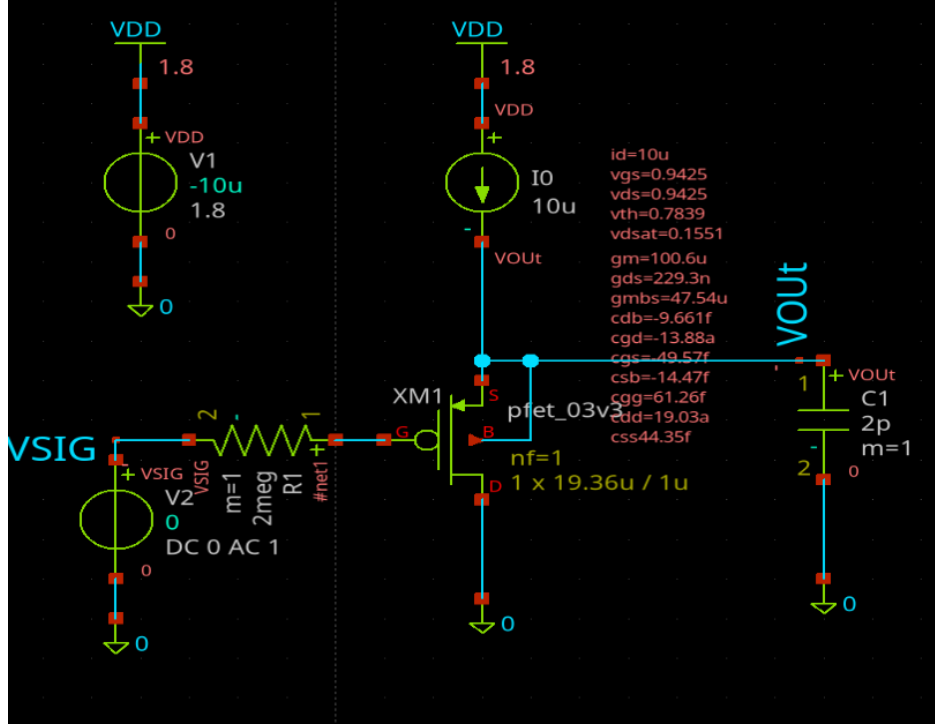


Figure 3: CD amplifier schematic with annotated OP parameters.

Parameter	Value
I_D	$10 \mu A$
V_{GS}	0.9425 V
V_{DS}	0.9425 V
V_{th}	0.7839 V
V_{DSAT}	0.1551 V
g_m	$100.6 \mu A/V$
g_{ds}	$229.3 nA/V$
g_{mb}	$47.54 \mu A/V$
C_{db}	9.661 fF
C_{gd}	13.88 aF
C_{gs}	49.57 fF
C_{sb}	14.47 fF

Table 2: OP simulation parameters.

Saturation check: $|V_{DS}| = 0.9425 \text{ V} > V_{DSAT} = 0.1551 \text{ V}$, so the transistor is well into saturation (also $|V_{GS}| - V_{th} = 0.9425 - 0.7839 = 0.1586 \text{ V}$, close to V_{DSAT} as expected). Good, so the OP is valid.

Since the source is tied to the body, $V_{SB} = 0$ at all times and even though g_{mb} is not zero, the small-signal body-effect current $g_{mb}v_{bs}$ is always zero because $v_{bs} = 0$ – that’s exactly why we use a PMOS in its own well for the CD amp, to kill the body effect that would otherwise degrade the gain.

2.2 AC Analysis

I ran an AC sweep from 1 Hz to 10 GHz, 20 points/decade.

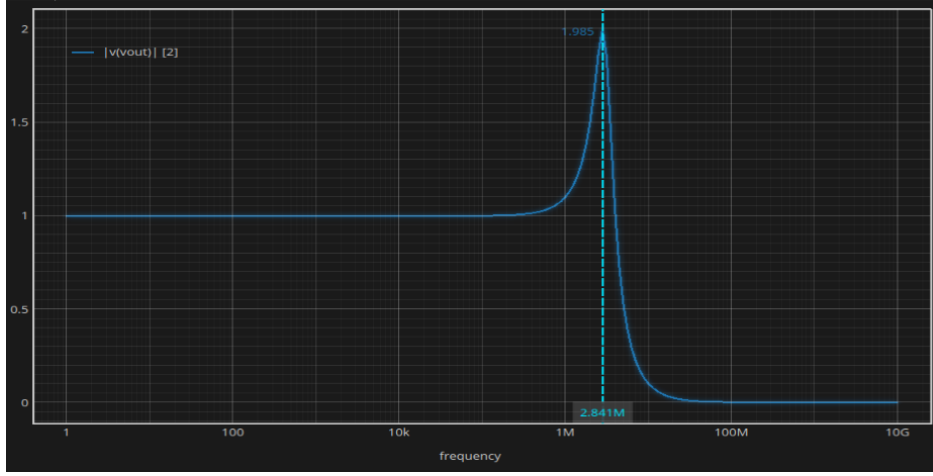


Figure 4: AC magnitude response, base case ($C_L = 2\text{pF}$, $R_{sig} = 2\text{M}\Omega$). Peak = 1.985 at $f = 2.841\text{ MHz}$.

There's a clear peaking around 2.84 MHz where the gain goes up to ≈ 1.985 (about +5.96 dB) even though the DC gain is basically 1. That's the frequency-domain peaking problem mentioned in the lab.

Hand analysis of the peaking

To understand where this peak comes from I derived the small-signal transfer function of the source follower including R_{sig} , C_{gs} , and C_L (I neglected C_{gd} since it's only $\approx 14\text{ aF}$, totally negligible next to $C_{gs} = 49.57\text{ fF}$, and I neglected the small g_{ds} terms that multiply $R_{sig}C_{gs}$ since $g_{ds} \ll g_m$).

Doing KCL at the gate node and the output node (gate node A after R_{sig} , output node = source of M1, drain grounded), I got:

$$\frac{V_{out}(s)}{V_{sig}} \approx \frac{1 + s \frac{C_{gs}}{g_m}}{1 + s \frac{C_{gs} + C_L}{g_m} + s^2 \frac{R_{sig}C_{gs}C_L}{g_m}}$$

Comparing the denominator to the standard 2nd order form $1 + \frac{s}{Q\omega_o} + \frac{s^2}{\omega_o^2}$:

$$\omega_o = \sqrt{\frac{g_m}{R_{sig}C_{gs}C_L}}, \quad Q = \frac{\sqrt{g_m R_{sig} C_{gs} C_L}}{C_{gs} + C_L}$$

Plugging in $g_m = 100.6\text{ }\mu\text{A/V}$, $C_{gs} = 49.57\text{ fF}$, $C_L = 2\text{ pF}$, $R_{sig} = 2\text{ M}\Omega$:

$$\omega_o = \sqrt{\frac{100.6 \times 10^{-6}}{(2 \times 10^6)(49.57 \times 10^{-15})(2 \times 10^{-12})}} = 2.252 \times 10^7 \text{ rad/s} \Rightarrow f_o \approx 3.585 \text{ MHz}$$

$$Q = \frac{\sqrt{(100.6 \times 10^{-6})(2 \times 10^6)(49.57 \times 10^{-15})(2 \times 10^{-12})}}{49.57 \times 10^{-15} + 2 \times 10^{-12}} \approx 2.18$$

Since $Q \approx 2.18 \gg 0.707$, the poles are complex and underdamped $\rightarrow$ the system *is* underdamped and we should expect a sharp resonant peak, which matches the simulation qualitatively (hand calc gives $f_o \approx 3.59$ MHz vs. simulated peak at 2.84 MHz – about 26% off, which I think is reasonable given I ignored C_{gd} , C_{db} , C_{sb} and the finite g_{ds} in the hand formula. The high Q correctly predicts the strong peaking we see).

C_L sweep: 2p, 4p, 8p

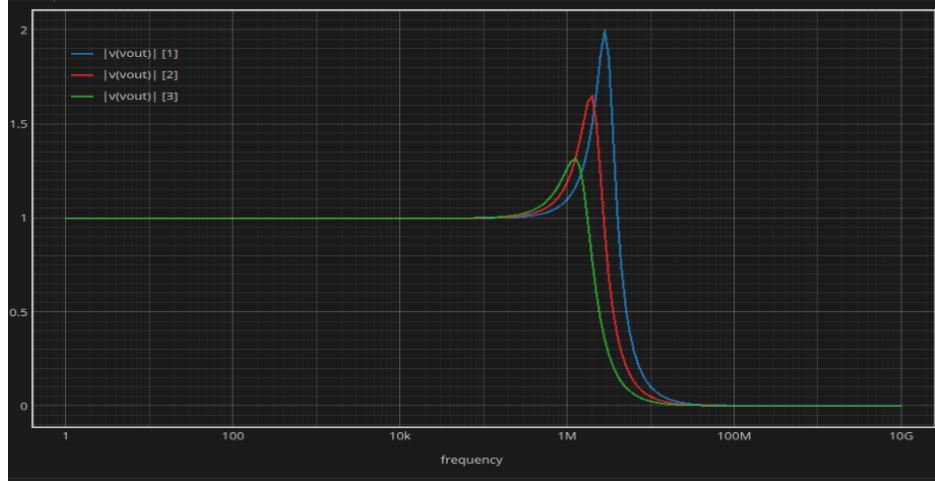


Figure 5: AC magnitude, $C_L = 2\text{p}$ (blue), 4p (red), 8p (green), $R_{sig} = 2\text{M}\Omega$.

```
No. of Data Rows : 201
binary raw file "cd_amp_cap.raw"
gain_peak_cap      = 2.385099e+00 at= 1.258925e+06
ngspice 1 -> █
```

Figure 6: ngspice measurement output for the C_L sweep (last iteration printed, $C_L = 8\text{p}$).

C_L	Peak gain (sim, approx.)	Hand-calc Q
2 pF	1.985 (5.96 dB)	2.18
4 pF	≈ 1.65 (≈ 4.4 dB, read off graph)	1.56
8 pF	2.305 dB $\Rightarrow V_{out} \approx 1.303$ (from meas, $f = 1.259$ MHz)	1.11

Table 3: Peaking vs. C_L .

So as C_L increases, the peaking gets smaller and the peak also moves to a lower frequency. This matches my Q formula perfectly – Q goes down as C_L goes up (since C_L is in the denominator's sum $C_{gs} + C_L$ growing faster than the numerator inside the square root). Physically, a bigger C_L pushes the output pole to a lower frequency, so it separates from the "negative resistance" effect of $R_{sig}-C_{gs}$ and the response becomes less resonant (closer to overdamped). This is basically the same trick suggested in the lab: separate the input and output poles to kill the peaking.

R_{sig} sweep: 20k, 200k, 2M

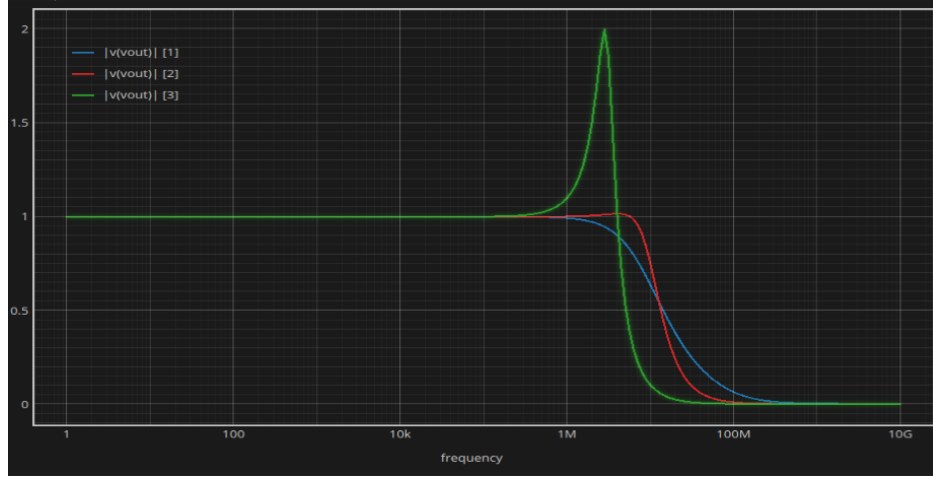


Figure 7: AC magnitude, $R_{sig} = 20k$ (blue), $200k$ (red), $2M\Omega$ (green), $C_L = 2pF$.

```
No. of Data Rows : 201
binary raw file "cd_amp_res.raw"
gain_peak_res      = 5.996595e+00 at= 2.818383e+06
Error: missing name for output.
ngspice 1 -> █
```

Figure 8: ngspice measurement output for the R_{sig} sweep (last iteration printed, $R_{sig} = 2M\Omega$).

R_{sig}	Peak gain	Hand-calc Q
20 k Ω	≈ 0 dB, basically flat, no real peak	0.22 (overdamped)
200 k Ω	very small bump, close to 0 dB	0.69 (borderline)
2 M Ω	5.997 dB $\Rightarrow V_{out} \approx 1.996$	2.18 (underdamped)

Table 4: Peaking vs. R_{sig} .

This makes total sense with the Q formula too: $Q \propto \sqrt{R_{sig}}$, so a small R_{sig} gives low Q (overdamped, no peaking, just a normal single-pole-ish rolloff) while a large R_{sig} pushes Q above 0.707 and you get the resonant peak. This is exactly the point of the lab – a source follower is "dangerous" if you drive it from a high-impedance source and load it with a cap, you can get significant gain peaking even though a source follower is "supposed" to just be a unity-gain buffer.

2.3 Transient Analysis

Pulse source: delay = $2\mu s$, initial = 0V, pulse = 100mV, rise/fall = 1ns, width = $4\mu s$, period = $8\mu s$. Transient run for $10\mu s$ with 10ns step.



Figure 9: V_{sig} and V_{out} vs. time, base case.

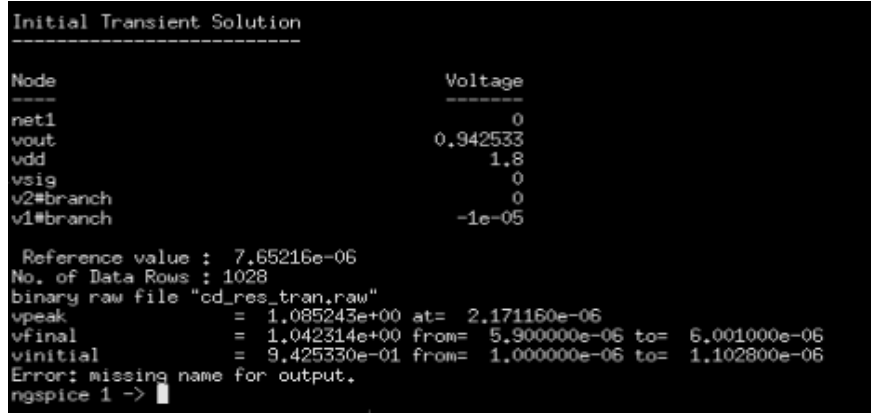


Figure 10: Initial transient solution and meas results (base case).

From the simulator: $V_{initial} = 0.942533$ V (settled value with $V_{sig} = 0$), $V_{final} = 1.042314$ V (settled value with $V_{sig} = 100\text{mV}$), $V_{peak} = 1.085243$ V at $t = 2.171\mu\text{s}$.

DC shift: With $V_{sig} = 0$, $V_{out} = 0.942533$ V, so

$$\Delta V_{DC} = V_{out} - V_{in} = 0.9425 - 0 = 0.9425 \text{ V}$$

and this is exactly equal to $|V_{GS}| = 0.9425$ V from the OP analysis. Makes sense: for a source follower $V_{out} = V_{in} - V_{GS}$ (magnitude-wise), so the DC shift is V_{GS} . Since this is a PMOS with the source above the drain, the output sits *above* the input by $|V_{GS}|$. If I wanted to shift the signal down instead of up, I would use an NMOS source follower instead (there $V_{out} = V_{in} - V_{GS}$ with $V_{GS} > 0$, so the output sits below the input).

Overshoot: using the formula from the hint,

$$\text{overshoot \%} = \frac{V_{peak} - V_{final}}{V_{final} - V_{initial}} \times 100 = \frac{1.085243 - 1.042314}{1.042314 - 0.942533} \times 100 \approx 43.0\%$$

this is the ringing problem caused by the underdamped 2nd-order response we calculated above ($Q \approx 2.18$).

Also as a check: the step size is $V_{final} - V_{initial} = 0.0998$ V for a 100mV input step, so the DC gain ≈ 0.998 , matching $A_{v0} = g_m/(g_m + g_{ds}) = 100.6/(100.6 + 0.229) = 0.9977$ from the OP values. Nice, consistent with the hand formula.

C_L sweep and R_{sig} sweep



Figure 11: V_{out} vs time, $C_L = 2p/4p/8p$ sweep.



Figure 12: V_{out} vs time, $R_{sig} = 20k/200k/2M\Omega$ sweep.

Same trend as the AC results: the case with the highest Q (smallest C_L , largest R_{sig} , i.e. the base case $C_L = 2p$, $R_{sig} = 2M\Omega$) shows the most overshoot/ringing, while increasing C_L or decreasing R_{sig} makes the response calmer and less oscillatory – exactly the same conclusion as the frequency domain, just seen in the time domain. Basically overshoot in time domain and peaking in frequency domain are two faces of the same underdamped 2nd order response, so it's not surprising they track each other with C_L and R_{sig} .

2.4 Z_{out} (Inductive Rise)

Here C_L was replaced with an AC current source (magnitude 1) and the AC input was removed, so $V_{out}/I_{test} = Z_{out}$.

```
Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 101
binary raw file "cd_amp_zout.raw"
zout_low_freq      = 9.917681e+03
zout_high_freq     = 1.881809e+05 at= 3.162278e+07
ngspice 1 ->
```

Figure 13: ngspice output: Z_{out} at low frequency and Z_{out} peak.

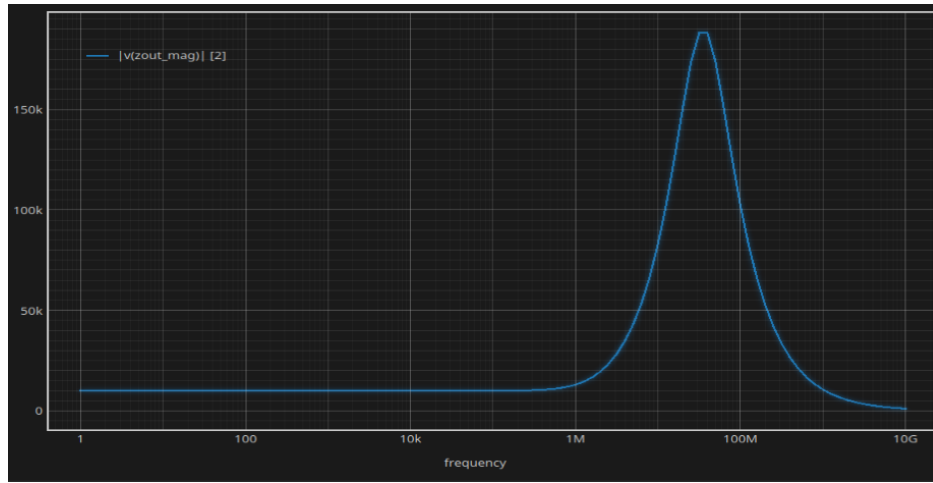


Figure 14: $|Z_{out}|$ vs frequency.

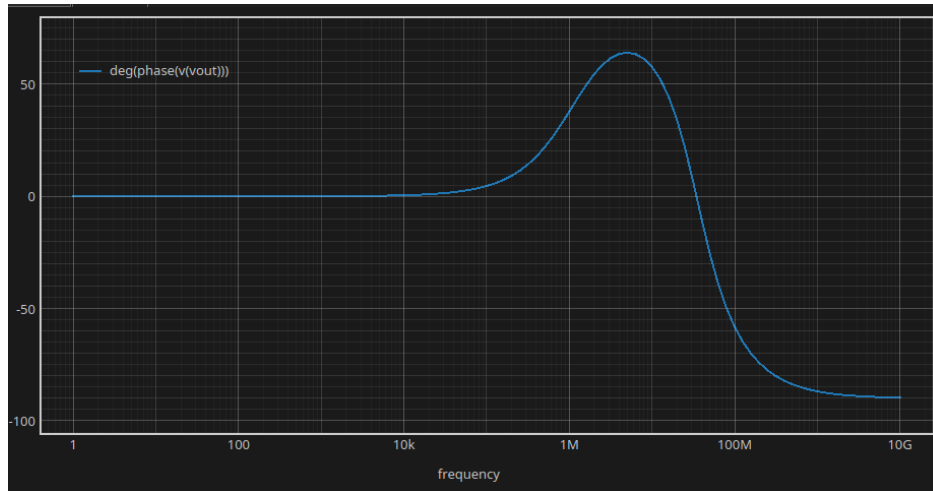


Figure 15: $\angle Z_{out}$ vs frequency.

From the simulation: $Z_{out}(1\text{kHz}) = 9917.68 \Omega$, and Z_{out} peaks at $188.18 \text{ k}\Omega$ around $f = 31.6 \text{ MHz}$.

Hand analysis

$$Z_{out}(s) = \frac{1 + sR_{sig}C_{gs}}{(g_m + g_{ds}) + sC_{gs}(1 + R_{sig}g_{ds})}$$

Low frequency: as $s \rightarrow 0$,

$$Z_{out}(0) = \frac{1}{g_m + g_{ds}} = \frac{1}{100.6\mu A/V + 0.229\mu A/V} = \frac{1}{100.83\mu A/V} \approx 9918 \Omega$$

This matches the simulated 9917.68Ω almost exactly (basically same number).

Zero: $1 + sR_{sig}C_{gs} = 0 \Rightarrow f_z = \frac{1}{2\pi R_{sig}C_{gs}} = \frac{1}{2\pi(2 \times 10^6)(49.57 \times 10^{-15})} \approx 1.605 \text{ MHz.}$

Pole: $(g_m + g_{ds}) + sC_{gs}(1 + R_{sig}g_{ds}) = 0 \Rightarrow f_p = \frac{g_m + g_{ds}}{2\pi C_{gs}(1 + R_{sig}g_{ds})}$. With $R_{sig}g_{ds} = (2 \times 10^6)(229.3 \times 10^{-9}) = 0.459$, $f_p \approx \frac{100.83\mu A/V}{2\pi(49.57\text{fF})(1.459)} \approx 222 \text{ MHz.}$

So the zero sits at $\approx 1.6 \text{ MHz}$, well below the pole at $\approx 222 \text{ MHz}$. Between these two frequencies the numerator $(1 + j\omega R_{sig}C_{gs})$ dominates the shape and $|Z_{out}|$ rises roughly linearly with frequency, i.e. it looks just like an inductor. In fact, in that mid-band region

$$Z_{out}(s) \approx \frac{sR_{sig}C_{gs}}{g_m + g_{ds}} = sL_{eq}, \quad L_{eq} = \frac{R_{sig}C_{gs}}{g_m + g_{ds}} \approx \frac{(2 \times 10^6)(49.57 \times 10^{-15})}{100.83 \times 10^{-6}} \approx 0.983 \text{ mH}$$

Checking this at the simulated peak frequency (31.6 MHz):

$$|Z_{out}| \approx \omega L_{eq} = 2\pi(31.6 \times 10^6)(0.983 \times 10^{-3}) \approx 195 \text{ k}\Omega$$

which is nicely close to the simulated peak of $188.18 \text{ k}\Omega$. So physically, the "inductive rise" happens because R_{sig} together with C_{gs} forms a kind of effective inductor looking into the source terminal – this is the classic result that a resistor in series with the gate of a source follower makes the output impedance look inductive.

Why does Z_{out} eventually fall at high frequency? My 1-zero/1-pole hand formula (ignoring C_{gd}) predicts the pole around 222 MHz, but the simulated peak is actually at 31.6 MHz – quite a bit lower. As the lab hint says, C_{gd} (even though it's only $\approx 14 \text{ aF}$) is in parallel with R_{sig} at the gate node, and at high enough frequency its impedance becomes small enough to start shorting the gate node to ground. That kills the boost from the R_{sig} - C_{gs} combination, so the effective "resistance driving the gate" drops and Z_{out} stops rising and comes back down. In reality there are also C_{db} and C_{sb} sitting on the output/drain nodes that I ignored in the simplified hand model (I assumed an ideal capacitor-free current source), and these extra parasitic capacitances also pull the real pole location down from my 222 MHz hand estimate to the simulated $\sim \text{MHz}$ -tens range. So the hand analysis gets the right physical picture (rise then fall, low-frequency value matches almost exactly, and even the peak magnitude estimate is in the right ballpark) but isn't 100% quantitatively exact for the high-frequency pole location, which is expected since I simplified the model.

Quantity	Hand calc	Simulation
Z_{out} @ low freq	9918 Ω	9917.68 Ω
Zero frequency	1.6 MHz	(phase peaks near $\sim 2 \text{ MHz}$, close)
Pole frequency (no C_{gd})	222 MHz	peak actually at 31.6 MHz ($C_{gd}/C_{db}/C_{sb}$ effect)
$ Z_{out} _{peak}$ (inductive estimate)	$\approx 195 \text{ k}\Omega$	188.18 $\text{k}\Omega$

Table 5: Hand analysis vs. simulation summary for Z_{out} .

The phase plot also supports this : phase rises up towards $\approx 63^\circ$ around 1–2 MHz (inductive-ish region, consistent with the zero I calculated), then dips down to a minimum near 60 MHz (this dip is from the extra high-frequency pole/zero pair that I neglected, i.e. $C_{gd}/C_{db}/C_{sb}$ effects), and then climbs back up toward 90° at very high frequency.

3 Part 2.5: Peaking/Ringing Fix (Bonus)

The straightforward fix mentioned in the lab, and the one I demonstrated with the sweeps above, is to keep the input pole (set by R_{sig} and C_{gs}) and the output pole (set by C_L/g_m) far apart from each other – either by lowering R_{sig} or by increasing C_L , both of which lower Q below 0.707 and remove the peaking. I did not implement the compensation network from Johns & Martin Section 4.4 for the bonus part, but conceptually it works by adding an element (typically a resistor in series with a compensation capacitor from gate to output, or similar) that cancels the negative real part of the input impedance that the source follower presents when driven by R_{sig} , which is really the root cause of the whole peaking/ringing behavior in the first place.